Team ICB floorplanner: code, build and reproduction guide

ICCAD 2026 CAD Contest, Problem C (FloorSet-Lite) | 10/18/2026

Latest results

Measure (contest evaluator, 100 validation designs)	Team ICB 10/18
Official weighted score (exp(blocks/12) weights)	1.0343
Ground-truth layouts, scored by the same evaluator	1.108
Published leader (10/6 leaderboard, raw cost)	1.084
Designs where we beat the ground truth's own score	76 / 100
Feasible layouts	100 / 100
Mean HPWL gap / area gap / relative violations	0.019 / 0.025 / 0.004
Mean runtime per design (8-core laptop, see 6)	641 s

Reference result file: iccad2026contest/official_eval_1018_restart.json (every design's layout, cost and runtime). 1.0 means the ground truth's wirelength and area; the ground truth itself scores 1.108 because it breaks some of its own edge and cluster constraints.

This document is for someone who wants to reproduce and verify these results: what to check out, how to install and run it, how to compare their numbers with ours, and how every trained model was produced.

1. Get the code

```
git clone git@github.com:andyfox-rushc/FloorSet.git
cd FloorSet
# all solver commands below run from FloorSet/iccad2026contest
```

The repository is a fork of Intel's FloorSet repository. The top level holds Intel's dataset loaders and the 100 validation designs (LiteTensorDataTest/); our solver lives in iccad2026contest/ next to the contest evaluator.

2. Install

No compilation step: everything is Python. Tested with Python 3.14.4 and the versions below (requirements.txt gives the minimum versions):

Package	Tested version
torch (CPU build is enough)	2.14.0
numpy	2.5.2
shapely	2.1.2
matplotlib	3.11.1
tqdm	4.70.0
requests	2.34.2

```
cd FloorSet
python3 -m venv venv
source venv/bin/activate
pip install -r iccad2026contest/requirements.txt
```

The scripts call ../venv/bin/python from iccad2026contest/, i.e. the venv inside FloorSet/. The validation data is in the repository; the 1M-design training set (about 15 GB download, 24 GB unpacked, into FloorSet/floorset_lite/) is only needed to retrain models (section 7) and downloads automatically on first use.

3. Run and verify

Full run (the official number)

```
cd iccad2026contest
../venv/bin/python iccad2026_evaluate.py --evaluate my_optimizer.py \
    --output my_run.json            # ~18 h on 8 cores
../venv/bin/python scripts/compare_with_gt.py my_run.json official_eval_1018_restart.json
```

The evaluator prints the total score; compare_with_gt.py scores the ground-truth layouts with the same evaluator and prints, for each result file, the weighted score, the mean gaps, how many designs beat the ground truth and the largest losses. All defaults are the configuration that produced 1.0343; no environment variables are needed. The count of designs that beat the ground truth moves more than the score between runs (76 in this run, 77 in the 10/17 run at 1.0404, 72 in the 10/16 run at 1.0432).

Quick check (about 15 + 35 minutes)

```
../venv/bin/python scripts/solver_ab.py --ids 0 98 --out quick.json
../venv/bin/python scripts/anchor_results.py --test-ids 0 49 98 --out-dir anchors
```

solver_ab.py solves chosen designs one at a time exactly as the evaluator does. Ours: design 0 1.000 and design 98 1.000 in the official run; 1.000-1.012 and 1.000 over three further runs (ground truth 1.091 and 1.080). Design 98 takes about 16 minutes, design 49 about 17 (it keeps a violation, so it always gets 3 solves). anchor_results.py also draws each layout next to the ground truth (PNG files).

What to expect

Results vary from run to run: the packer searches for a fixed wall-clock time, so machine speed and load change how far it gets. Our repeated runs of one configuration differ by about +-0.01 on the weighted score, and single designs much more (design 49: 1.127-1.274 over our runs). Run on an otherwise idle machine; a slower or busy machine scores worse. Six worker processes are used (FLOORSET_PROCS, default min(6, cpus)).

4. Files needed

Paths relative to FloorSet/. 'Intel' = unchanged from the upstream repository.

Solver (run time)

File	Role
iccad2026contest/my_optimizer.py	Entry point (MyOptimizer.solve), the whole pipeline
iccad2026contest/iccad2026_evaluate.py	Contest evaluator (Intel, unchanged)
iccad2026contest/rl/__init__.py	Package marker
iccad2026contest/rl/data.py	Contest inputs -> FloorplanInstance
iccad2026contest/rl/ordering.py	Placement order for the RL policy
iccad2026contest/rl/env.py	Grid placement environment (hard constraints)
iccad2026contest/rl/encoder.py	Graph network over the netlist
iccad2026contest/rl/networks.py	RL policy / value networks
iccad2026contest/rl/ppo.py	PPO rollouts and updates
iccad2026contest/rl/reward.py	Contest cost as the RL reward
iccad2026contest/rl/finetune.py	Per-design PPO fine-tuning of the policy
iccad2026contest/rl/checkpoint_meta.py	Checkpoint metadata (canvas utilization)
iccad2026contest/rl/compaction.py	Gravity compaction
iccad2026contest/rl/anneal.py	Annealing polish
iccad2026contest/rl/repack.py	Re-packing polish, exact cost helpers
iccad2026contest/rl/slicing.py	Slicing polish
iccad2026contest/rl/legalize.py	Wall repair / legalization
iccad2026contest/rl/analytic.py	Analytic placers, learned candidates, best-of
iccad2026contest/rl/learned_placer.py	Learned placer network (centres, relations)
iccad2026contest/rl/sizing.py	Differentiable shape fitting
iccad2026contest/rl/packer.py	Sequence-pair packer, squeeze, wirelength relaxation
cost.py utils.py visualize.py	Intel helpers imported by the evaluator
liteLoader.py lite_dataset.py litetestLoader.py	Intel data loaders
lite_dataset_test.py prime_dataset.py	Intel data loaders
LiteTensorDataTest/ (200 files)	Intel: the 100 validation designs + labels

Trained models (iccad2026contest/checkpoints/)

File	Size	Role
policy.pt + policy.pt.meta.json	304 KB	RL placement policy (PPO), canvas utilization 0.70
placer_a.pt	1.3 MB	Learned placer A: block centres + aspect ratios
placer_r.pt	7.4 MB	Relation network 1 (pairwise relations + centres)
placer_relB.pt	7.4 MB	Relation network 2 (independent training run)

Training, verification and documentation

File	Role
iccad2026contest/rl/train.py, rl/pretrain.py	RL policy training (python -m rl.train)
iccad2026contest/scripts/train_placer.py	Learned placer / relation network training
iccad2026contest/scripts/polish_bench.py	Shared evaluator setup used by the scripts
iccad2026contest/scripts/__init__.py	Package marker
iccad2026contest/scripts/compare_with_gt.py	Verify a result file against the ground truth
iccad2026contest/scripts/solver_ab.py	Solve chosen designs one at a time
iccad2026contest/scripts/anchor_results.py	Solve + draw layouts vs ground truth
iccad2026contest/scripts/big_designs_pdf.py	Ground truth vs ours for the largest designs (PDF)
iccad2026contest/scripts/generate_code_doc_pdf.py	Builds this document
iccad2026contest/official_eval_1018_restart.json	Reference result (1.0343)
iccad2026contest/algorithm.md	Design notes and every experiment, dated
iccad2026contest/requirements.txt	Python packages

We checked this list by copying only these files into an empty directory and running the solver, the placer training and the policy training from there.

5. How the solver works

For each design my_optimizer.py builds candidate layouts, packs the most promising ones with a long search, and returns the cheapest by the exact contest cost:

Stage	What it does
1. RL policy	policy.pt places blocks one at a time on a grid (GNN encoder + CNN position head), fine-tuned on the design with PPO for a few seconds ('base')
2. Analytic placers	Five wirelength-driven global placements from scratch at canvas utilizations 0.75-1.0 ('scratch...' candidates)
3. Learned placers	Networks trained on 1M ground-truth layouts predict block positions and pairwise left/right/below/above relations ('learned1-4')
4. Shape fitting	Gradient descent on soft-block aspect ratios for a fixed arrangement
5. Best-of	All candidates scored; the best plus the four learned ones go on
6. Packer	Sequence-pair annealing on the exact contest cost, 200 s at 77 blocks (scaled by (blocks/77)^2), one process per layout, shape refit halfway; split clusters penalized in the search and repaired exactly afterwards
7. Second round	The winner is packed once more from where it ended, with a search penalty for running past the outline revealed by preplaced edge blocks; kept only if cheaper (new 10/17)
8. Finish	Squeeze, wirelength relaxation, then the cheapest layout inside the box where the pins sit (preferred within 5% of the best cost); soft blocks up to 16:1 throughout
9. Restarts	While the cheapest layout still breaks a soft constraint, stages 1-8 run again with new seeds (at most 3 runs), the reruns also aiming for the outline; cheapest kept (new 10/18)

The policy and models used

Candidate	Checkpoint	Arrangement from
base	policy.pt	RL policy
scratch0.75 ... scratch1.0	(none)	analytic placement
learned	placer_a.pt	predicted centres
learned2	placer_r.pt	predicted pairwise relations
learned3	placer_r.pt	predicted centres
learned4	placer_relB.pt	predicted pairwise relations

Settings (defaults = the 1.0343 run)

Environment variable	Default	Meaning
FLOORSET_PACK	200	Packer seconds per layout at 77 blocks
FLOORSET_PROCS	min(6, cpus)	Worker processes
FLOORSET_MAX_ASPECT	16	Longest soft-block side / shortest (was 8 until 10/15)
FLOORSET_REATTACH	1	Exact repair of split clusters after packing (new 10/16)

FLOORSET_PACK_GAP	1	Search penalty on cluster-piece distance (new 10/16)
FLOORSET_REPACK_WINNER	1	Second packer round from the winner, kept if cheaper (new 10/17)
FLOORSET_WALL_PASS	100	Search penalty past preplaced walls in that round (new 10/17)
FLOORSET_SOLVE_RUNS	3	Most whole solves per design (new 10/18)
FLOORSET_RESTART_VIOL	1	Solve again only while soft violations remain (new 10/18)
FLOORSET_CANDIDATES	base,scratch0.75..1.0,learned..learned4	Candidate layouts
FLOORSET_CHECKPOINT	checkpoints/policy.pt	RL policy
FLOORSET_PLACER, ..._PLACER4	see table above	Learned placer checkpoints

algorithm.md documents every other switch, with the measurement that set its default.

6. Machine used for the results

Intel Core Ultra 7 258V (8 cores), 16 GB RAM, Windows 11 + WSL2 (Linux 6.18), CPU only. One official run of the 100 designs took about 18 hours (641 s per design on average, most of it the packer's fixed time budget).

7. Retraining the models (optional)

The checkpoints in the repository are the ones that produced the results. To retrain them, run the commands below from iccad2026contest/. Training stops on a wall-clock limit and runs on several threads, so a retrained model will not be bit-identical; expect the solver's score to land within the run-to-run noise rather than exactly. Judge a retrained model by a full evaluation run (section 3), not by its validation error: training network 1 on to 358k designs lowered that error but scored worse (1.0709).

Training data

```
../venv/bin/python -c "from iccad2026_evaluate import get_training_dataloader; \
    get_training_dataloader(data_path='../', batch_size=1)"   # -> ../floorset_lite/
```

RL policy: checkpoints/policy.pt

```
../venv/bin/python -m rl.train --iterations 4000 --episodes-per-iter 4 \
    --ppo-epochs 4 --checkpoint checkpoints/policy.pt --checkpoint-every 500 \
    --threads 3 --data-seed 12345
```

From scratch: reward-prediction pretraining of the encoder, then PPO over the training designs with an adaptive canvas utilization (settles at 0.70). We use the iteration-4000 checkpoint (13.4 hours on 3 threads); longer training did not improve the pipeline.

Learned placer A: checkpoints/placer_a.pt (128 wide, 6 layers)

```
../venv/bin/python scripts/train_placer.py --hours 7 --threads 2 --val-every 10000 \
    --out checkpoints/placer.pt
```

placer_a.pt is the snapshot after about 290k training designs (validation centre error 0.067), taken while the run continued; to match it, stop near 290k designs.

Base for the relation networks: placer_d.pt (256 wide, 8 layers)

```
../venv/bin/python scripts/train_placer.py --hours 6 --threads 4 --seed 4 --lr 5e-4 \
    --hidden 256 --layers 8 --norm --val-every 5000 --out checkpoints/placer_d.pt
```

Relation network 1: checkpoints/placer_r.pt (two stages)

```
../venv/bin/python scripts/train_placer.py --hours 5 --threads 4 --seed 5 --lr 3e-4 \
    --hidden 256 --layers 8 --norm --rel --resume checkpoints/placer_d.pt \
    --val-every 5000 --out checkpoints/placer_rel.pt
../venv/bin/python scripts/train_placer.py --hours 6 --threads 6 --seed 6 --lr 2e-4 \
    --hidden 256 --layers 8 --norm --rel --resume checkpoints/placer_rel.pt \
    --val-every 5000 --out checkpoints/placer_r.pt
```

About 312k designs in total; validation centre error 0.050, relation accuracy 0.918.

Relation network 2: checkpoints/placer_relB.pt (independent of network 1)

```
../venv/bin/python scripts/train_placer.py --hours 8 --threads 4 --seed 31 --lr 2e-4 \
    --hidden 256 --layers 8 --norm --rel --resume checkpoints/placer_d.pt \
    --val-every 5000 --out checkpoints/placer_relB.pt
```

About 357k designs; validation centre error 0.050, relation accuracy 0.917.